LESSON 6.6

Identify Equivalent Algebraic Expressions

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can show that two expressions are equivalent by simplifying and combining like terms.

WHY THIS MATTERS At the music studio, two producers wrote formulas for the cost of renting equipment. Producer A wrote 3x + 2x + 10, and Producer B wrote 5x + 10. They look different, but are they really? Let's check: if x = 4, Producer A gets 3(4) + 2(4) + 10 = 30, and Producer B gets 5(4) + 10 = 30. Same result — they're equivalent expressions!

REMEMBER

Identify Equivalent Algebraic Expressions

Equivalent Algebraic Expressions Two expressions are equivalent when they always give the same value, no matter what number you put in for the variable. They can look different but mean the same thing.

1. Equivalent expressions produce the same value for any substitution of the variable.
2. Simplify expressions using algebraic properties (Distributive, Commutative, Associative).
3. Test equivalence by substituting two different test numbers

Words: Equivalent Expressions *(Expresiones equivalentes)* · Equivalent *(Equivalente)* · Simplify *(Simplificar)* · Like Terms *(Términos semejantes)* · Combine *(Combinar)*

PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

Which expression means '7 more than a number x'?

(A) x + 7
(B) 7x
(C) x − 7
(D) 7 − x

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

A ticket costs $9. Which expression gives the cost of t tickets?

(A) 9t
(B) t + 9
(C) 9 + t
(D) t ÷ 9

SHOW YOUR WORK

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

Evaluate 4n + 3 when n = 5.

(A) 23
(B) 27
(C) 35
(D) 45

My work — one step per line	What I did to both sides

1. **Look** *What is done to the variable?*
2. **Undo** *Do the opposite — both sides.*
3. **Simplify** *One side at a time.*
4. **Check** *Put it back in.*

4 **Circle the letter of the best answer. Show how you know.**

Evaluate 30 − 4y when y = 6.

(A) 6
(B) 20
(C) 24
(D) 156

My work — one step per line	What I did to both sides

1. **Look** *What is done to the variable?*
2. **Undo** *Do the opposite — both sides.*
3. **Simplify** *One side at a time.*
4. **Check** *Put it back in.*

5 **Circle the letter of the best answer. Show how you know.**

A music studio charges different clients using different formulas. Client A pays 4h + 2h + 30 for h hours. Client B pays 6h + 30. A new client asks if the pricing is fair. Simplify 4h + 2h + 30.

(A) 4h + 32
(B) 6h + 30
(C) 6h + 30h
(D) 36h

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

A music studio charges different clients using different formulas. Client A pays 4h + 2h + 30 for h hours. Client B pays 6h + 30. A new client asks if the pricing is fair. Are the two pricing formulas equivalent?

(A) Yes — 4h + 2h + 30 simplifies to exactly 6h + 30
(B) No — Client A pays more
(C) No — Client B pays more
(D) Only when h = 0

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How did you decide which expressions belong in the same group? What strategies did you use?

SENTENCE STARTER I knew ___ and ___ are equivalent because when I ___ the terms, I got the same expression.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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1 **Circle the letter of the best answer. Show how you know.**

A music studio charges different clients using different formulas. Client A pays 4h + 2h + 30 for h hours. Client B pays 6h + 30. A new client asks if the pricing is fair. Test it: what does each expression equal when h = 5?

(A) A = 50 and B = 60
(B) Both equal 60
(C) A = 60 and B = 55
(D) Both equal 90

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Producer A wrote 2(x + 3) for a beat's cost. Which expression is equivalent?

(A) 2x + 6
(B) 2x + 3
(C) 2x + 5
(D) x + 6

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

A track stacks 3x and then one more x. Which expression is equivalent to 3x + x?

(A) $3x^2$
(B) 3x
(C) 4x
(D) 4

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Producer B flipped a formula to 5 + 2x. Which expression is equivalent?

(A) 2x + 5
(B) 5x + 2
(C) 7x
(D) 10x

SHOW YOUR WORK

5 **Write the name of the correct group on each line.**

The target beat formula is 4x + 6. Spin each expression onto the deck where it belongs: equivalent to 4x + 6, or NOT equivalent.

Groups: Equivalent to 4x + 6 | NOT Equivalent

_____ 2(2x + 3)
_____ 4x + 3 + 3
_____ x + x + x + x + 6
_____ 4x + 10
_____ 10x
_____ 4(x + 6)

6 **Write the letter of the matching item on each line.**

Match each studio beat formula to its equivalent remix.

_____ 3(x + 4)
_____ 7y + y
_____ 6 + 5n
_____ 2(4a − 1)
_____ 9m − 3m

(A) 3x + 12
(B) 8y
(C) 5n + 6
(D) 8a − 2
(E) 6m

EXPLAIN YOUR THINKING

How did you decide which expressions belong in the same group? What strategies did you use?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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